

Abstracting Causal Models

Definition 3.12

Set $V \rightarrow$ endogenous variables

$$\vec{x} \subseteq V, \vec{x} \in R(\vec{x})$$

let $Rst(V, \vec{x}) = \{ \vec{v} \in R(V) : \vec{x} \text{ is the restriction of } \vec{v} \text{ to } \vec{x} \}$.

Given $T: R(V_L) \rightarrow R(V_H)$, define $w_T(\vec{X} \leftarrow \vec{x}) = \vec{y} \leftarrow \vec{y}$ if there exist $\vec{y} \subseteq V_H$ (set of high level variables which are endogenous) & $\vec{y} \in R(\vec{y})$ s.t.:

$$T(Rst(V_L, \vec{x})) = Rst(V_H, \vec{y})$$

consistency check to ensure that a specific action you take in a detailed micro model maps perfectly to a single clear action in a simplified macro model. $\Rightarrow$ induced intervention mapping.

$$\tau(T) = \{ \tau(\vec{v}_L) : \vec{v}_L \in T \}, T \subseteq R(V_L) *$$

- T is a restriction set, a cloud of all possible low level states consistent with a low level intervention [T is a subset of low level variable ranges]

- $\tau(T) \rightarrow$ set of all high level states when applying abstraction function to every state in the low level model.

- Given $\vec{X} \leftarrow \vec{x}$ we have at most one such $\vec{y} \leftarrow \vec{y}$: if a low level intervention maps to a high level one, it can only map to exactly one high level intervention

- In other words, a single I_L cannot mean or map to 2 different I_H .

- I_L^T : Allowed low level interventions

- I_H^T : Allowed high level interventions

- if τ is surjective $\Rightarrow w_T(V_L \leftarrow \vec{v}_L) = V_H \leftarrow \tau(\vec{v}_L)$